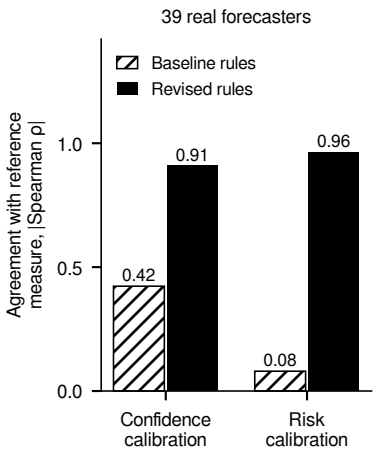
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Testing What a Score Measures: Validating and Repairing the Calibration Scores of a Personal Decision Analytics System

Shakil Ur Rehman*



Two of four baseline scores of a personal decision-analytics application computed their formulas correctly yet did not measure what their names claimed. Metamorphic relations derived from score names, simulation with known miscalibration and a replay of 4,793 real forecasts exposed both defects without a user study, alongside a race condition, unescaped output and an N+1 query pattern. The revised rules raise agreement with standard calibration measures from $|\rho| = 0.42$ and 0.08 to 0.91 and 0.96.

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